TISHA SEHRAWAT

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Profile

IBDP Student (M27) | Medicine, Neuroscience & Oncology | Bioinformatics & Healthcare Technology

Interested in how research and technology can shape the way we understand and treat disease. My work spans cancer research, neurodegenerative disease, bioinformatics, and healthcare technology, alongside clinical exposure across hospital departments.

Education

Intended Major: Biological Sciences (U.S.) | Medicine (U.K.)				
Grade	**School**	**Board**	**Year**	**Result**
Grade 12	10X International School, Bangalore	IBDP, Irregular Diploma (3 sciences)	2027	42/45 Predicted Biology HL - 7, Chemistry HL - 7, Math AA HL - 6, Physics SL - 7, English A L&L SL - 7, Hindi B SL - 6
Grade 11	10X International School, Bangalore	IBDP, Non-regular Diploma (3 sciences)	2026	FE: 36/42 — Biology HL (6), Chemistry HL (6), Math AA HL (5), Physics SL (6), Hindi B SL (6), English A L&L (7)
Grade 10	10X International School, Bangalore	IBMYP (10 subjects)	2025	50/56 (/7) — Biology (7), Chemistry (6), Physics (6), History (7), Design (7), Personal Project (7), English (6), Hindi (6), Extended Math (5), IDU (5)
Grade 9	10X International School, Bangalore	IBMYP (10 subjects)	2024	47/56

Standardised Test Scores

Test	Date Taken	Score	Section Breakdown
SAT	2025	1500 / 1600 (Superscore)	770 Math, 730 Reading & Writing
UCAT	Aug 29, 2026	1940, B3	VR – 600, DM – 700, QR – 640, SJ – Band 3
IELTS	Planned for October	—	—

Co-Curricular Activities & Leadership

House Captain, Student Council — 10X International School, Bangalore

Grade 11–12 | Oct 25 – Oct 26

- Lead and coordinate house-wide academic, cultural, and athletic initiatives; represent student interests and organize inter-house events.
- Assign and mentor house members; support peer mentorship programs.

Co-President, Science Club 'The Significant Figures' — 10X International School

Grade 10–11 | Aug 25 – May 26

- Co-founded and lead a student platform with over 30+ members, promoting scientific inquiry; organize workshops, discussions, and peer-led learning initiatives.

Founder & Editor-in-Chief, The Scientific Trail — 10X International School

Grade 11–12 | May 26 – Aug 26

- Founded and lead the school's first student-led science magazine, an official school publication engaging 60+ students in scientific writing, research, and communication.
- Develop and edit accessible science content spanning biomedical research, emerging technologies, and current scientific developments.

Editorial Team, 10Xpress (School Newspaper)

Grade 9–10 | Jan 24 – Jan 25

- Writing and editing analytical and narrative-driven pieces on health and medicine topics.

Editor, Yearbook Committee — 10X International School

Grade 9–10 | Jan 24 – Jan 25

- Lead editorial planning, content curation, and narrative design for the annual school publication.

TEDx Speaker — TEDx10X International School Youth

Grade 11 | Oct 25 – Jan 26

- Selected through a competitive curation process to deliver "Fear Is the First Symptom," exploring psychological barriers in healthcare and proposing human-centered medical innovation frameworks.

Member, International Youth Neuroscience Association

Grade 10–11 | Jan 25 – May 26

- Contribute to global neuroscience outreach initiatives and youth-led scientific engagement programs.

Competitions, Olympiads & Challenges

CREST Gold Award — Gold Medalist

Grade 12 | 2026

- Awarded the highest level of the internationally recognised CREST Awards programme administered by the British Science Association for an extended independent STEM research project assessed by industry professionals, for the project PlastiXcan.
- The research involved end-to-end project planning, experimental design, prototype development, data analysis, validation, and critical evaluation over more than 70 hours of independent research.
- The project received full marks (100%), demonstrating excellence in scientific methodology, innovation, technical execution, and communication.

International Chemistry Competition (IChC) — Finalist

Grade 11 | 2026

- Qualified for the Final Round after clearing the Qualification and Semi-Final Rounds; assessed on advanced chemistry problem-solving beyond the school curriculum.

International Biology Bowl (IBB) — Bronze Medalist (3rd Place)

Grade 11 | 2026

- Recognized across molecular biology, genetics, ecology, physiology, and evolution; awarded a USD 50 cash prize.

Diamond Challenge — Global Semi-Finalist

Grade 11 | 2026

- Advanced to the global Pitching Round with Team VisionXcelerate; built a 15-slide investor-ready pitch deck and delivered a 5-minute pitch evaluated by industry experts.

Pulitzer Center Local Letters for Global Change — Global Finalist (Global Health)

Grade 11 | 2026

- Selected from nearly 1,500 global submissions for a piece on pharmaceutical quality and health equity; published on the Pulitzer Center platform with a $75 civic engagement grant.

Yale International Relations Association (YIRA) — Honorable Mention

Grade 10 | 2025

- Recognized for analytical and writing skills addressing 'What's the biggest issue in the world today?', mentored by Yale undergraduates.

InnovateXHarrow Business Pitch Challenge — Runner-Up

Grade 9 | 2024

- Placed 2nd with BioBeats, a concept for biodegradable single-use headphones; served as Product Designer selecting sustainable materials.

NYAS Junior Academy Wearables Challenge — Global Winner (1st of 167 teams)

Grade 9 | 2024

- Part of Team VisionXcelerate, which won 1st place (top ~0.6%) for an AI/AR-driven assistive device for dementia patients; invited to present at the NYAS Virtual Student Symposium.

Stonehill Science Bowl — Runner-Up, Modern Problems Category

Grade 9 | 2023

- Placed 2nd with Team experiMENTAL, presenting a wildfire-mitigation solution.

National Astronomy Challenge (NAC) — Gold Medalist, National Rank 32

Grade 9 | 2023

- Ranked 32nd nationally for excellence in astrophysics problem-solving and analytical reasoning.

Harvard Model United Nations — Delegate

Grade 9 | Aug 23

- Represented Mexico in the Special Political & Decolonization Committee (SPECPOL).

Community Service & Social Impact

Co-Founder — Shiksham: Empowering Rural Girls Through Science, Mentorship & Holistic Learning

Oct 2023 – May 2025

- Co-founded a social-impact initiative providing STEAM education and holistic support for underserved rural girls in India.
- Taught 70+ rural girls across villages; led STEM workshops and mentorship sessions; expanded reach through NGO partnerships.
- Secured and channelled over ₹10,000 in funding toward books, stationery, and academic resources; managed tutoring programs and mental-wellness support.

Patient Intake & Care Observation — NGO Community Clinic

Jan 2026 – Apr 2026

- Assisted with patient intake and observed primary healthcare delivery for underserved populations; gained insight into public health accessibility and community-based care models.

Internships & Work Experience

Scientific Enrichment Month (SEM) Research Scholar — Columbia University Irving Medical Center

Jun 2026 – Jul 2026

- Selected for Columbia's academic enrichment program in cancer research, oncology, and biomedical innovation.
- Attend expert-led sessions on cancer biology, translational medicine, precision medicine, and pathways in academic medicine.

Bioinformatics & Evolutionary Genomics Intern — Pharmint Pte Ltd, Singapore

Jun 2025 – Jul 2025

Top-10 global pharma/biotech research organization

- Selected as 1 of 20 out of 500 applicants (~4% acceptance rate); analyzed genotype–phenotype correlations in beak morphology evolution across 30+ avian species using two developmental genes (ALX1, HMGA2).
- Applied multiple sequence alignment, phylogenetic reconstruction, and mutation mapping; processed 5,000+ genetic sequences and built 1,000+ evolutionary trees with ~95% accuracy in identifying conserved genetic regions.
- Identified 8+ significant mutations and 3 clear cases of convergent evolution across unrelated species; improved sequence-curation reliability by 75%; explored whether man-made environmental pressures (e.g., deforestation) drive genetic change.

Project Intern — SankalpTaru Foundation (CROASIS)

Jun 2025 – Sep 2025

- Designed and prototyped an internal Carbon Credit Tracking Dashboard for climate-data transparency; built responsive Figma UI components for credit estimation and emissions monitoring.
- Authored feature documentation and a user guide; contributed to the development roadmap, including vendor research and cost analysis.

AI Student Researcher — Inspirit AI *Oct 2024 – Dec 2024*

- Completed the AI Scholars Winter Program under mentorship from Stanford and MIT graduates.
- Built ML models to trace COVID-19 strain origins in the 'AI + Healthcare: DNA Detectives' module, applying dimensionality reduction to biological data.

Medical Research Scholar — Harvard Student Agencies (Future Doctors I) *May 2024 – Jul 2024*

- Completed coursework in genetics, immunology, endocrinology, and organic chemistry.
- Engaged in case-based clinical simulations analyzing patient data, differential diagnosis, and treatment planning.

Summer Researcher — ICREST Academy *Apr 2024 – Jul 2024*

- Completed ALP 101 and the six-week ALP 201 program covering microbiology, biochemistry, immunology, molecular biology, genetic engineering, and bioinformatics.

Scientific Researcher (Biology Track) — Science Mentorship Institute *Jul 2024 – Aug 2024*

- Completed the Biology Mentorship Program (BMP); conducted an independent research project and presented peer-reviewed papers in weekly journal clubs.

Research & Extended Projects

Machine Learning Classification of Cancer Types Using 5-Hydroxymethylcytosine (5hmC) Signatures

Jun 2024 – Ongoing

Centre for Fundamental Research and Creative Education (CFRCE) — Advanced Program Research Scholar, mentored by Dr. Robert Austin (only high-school researcher in the program)

- Developed a supervised ML classification framework integrating 5hmC epigenetic signatures from cfDNA sequencing to differentiate cancer types and progression stages, building a non-invasive early-detection diagnostic tool.
- Analyzed 72 variances across 1,000+ patient records; trained logistic regression, random forest, and gradient boosting models achieving 90–92% classification accuracy.
- Established a strong negative correlation between 5hmC biomarker depletion and cancer progression, demonstrating transformative potential for early detection in resource-limited healthcare settings.
- Work independently validated by a Stanford PhD and IISc faculty; attracted seed-stage interest from a NYC venture partner connected to 2A Biosciences; collaborated with a Princeton Professor of Biophysics and was mentored by a Princeton PhD.
- Achieved a 3.98/4.0 CGPA in the program with Outstanding evaluations in research aptitude and academic writing; submitted to the IRIS National Fair 2025; recommended for Year 2 continuation.

Prion Disease in India: A National Review & Diagnostics Analysis *Jul 2025 – Jul 2026*

Team Pangea — Independent Researcher & Published Author, Oxford Journal of Student Scholarship | mentored under an IIT Bombay PhD researcher

- Authored the first comprehensive national review of prion disease diagnostics in India, a critically underdocumented and neglected disease landscape, synthesizing 50+ peer-reviewed studies.
- Proposed AI-assisted detection frameworks and RNA-based therapeutic pathways, and designed a neurologist questionnaire to gather primary data on clinical prion cases in India.
- Published in a globally peer-reviewed journal with under 20% acceptance, reviewed by graduate scholars and early-career academics across institutions worldwide.
- Proposed an integrative framework for India combining genomic context, institutional partnerships, and precision-medicine policy investment.

Project TRIAD: The Roots of Intersecting African Diseases *Feb 2026 – Apr 2026*

Non-Trivial — Research Foundations Scholar (acceptance rate <1%)

- Designed an interdisciplinary global health research proposal on the convergence of sickle cell anaemia, malaria, and neurodegeneration across sub-Saharan Africa and South Asia.
- Developed a conceptual framework linking HbS allele frequency, malaria transmission intensity, and metabolic dysfunction pathways to long-term neurological outcomes.

- Proposed geospatial epidemiological modeling using IHME and Malaria Atlas Project datasets, and a systematic review protocol on malaria outcomes in SCA patients.

Publications

Prion Disease in India: A National Review & Diagnostics Analysis

Feb – Jul 2026

Published Author, Oxford Journal of Student Scholarship

- Authored a national review of prion disease diagnostics in India, synthesizing 50+ peer-reviewed studies and proposing AI-assisted diagnostic and precision-medicine frameworks.
- DOI: https://doi.org/10.65161/rec45REQuN6IdiomV

Machine Learning Classification of Cancer Types Using 5-Hydroxymethylcytosine (5hmC) Signatures

2026

Researcher, CFRCE

- Developed a machine-learning framework using cfDNA 5hmC signatures for non-invasive cancer classification and early detection; currently in the editing process at the Current Journal of Applied Science and Technology.
- Pre-print DOI: https://dx.doi.org/10.21203/rs.3.rs-10868632/v1

PlastiXcan: System and Method for Non-Destructive Microplastic Identification in Food and Water Using Controlled Optical Imaging and Edge-Based Artificial Intelligence

Published Aug 2026

Patent Holder & Inventor | Indian Patent Application No. IN202641095036 A1

- Developed a low-cost optical and AI-based system for identifying and quantifying microplastics in food and water; patent published by the Indian Patent Office. Submitted to the IRIS National Fair 2026.
- Published Patent, Indian Patenting Journal (p. 808): https://search.ipindia.gov.in/IPOJournal/Journal/ViewJournal
- Published News Article https://www.cioandleader.com/teen-inventor-builds-a-rs-4000-device-to-catch-the-plastic-we-cant-see/

Project CARE: Clinical Anxiety Reduction & Emotional Support System

Published Sep 2026

- A patient-centered healthcare innovation project translating clinical observations and behavioral research into a digital support system for hospital environments. Project CARE is a web application designed to reduce fear and uncertainty during high-stress healthcare moments through structured emotional check-ins, context-specific explanations of hospital workflows, and evidence-informed coping interventions.
- Built around a three-stage framework, Capture Fear, Clarify Uncertainty, and Create Comfort, the platform anticipates common patient and caregiver anxieties during diagnosis, surgery, consultations, and waiting periods while supporting communication, emotional regulation, and informed engagement without replacing clinical decision-making.
- Published App on the Play Store in 3 languages (English, Hindi, Kannada): https://play.google.com/store/apps/details?id=com.projectcare.app&hl=en_US

Innovation, Maker & Design Projects

PlastiXcan — Founder, Lead Researcher & Patent Holder | Low-Cost Microplastic Detection Device

Nov 2025 – Ongoing

Patenting and deployment in households, with validation from FSSAI, the Indian Government, and CREST Gold Awards

- Built a portable microplastic detection system combining Raspberry Pi imaging, OpenCV computer vision, and optional ML (TensorFlow Lite) to detect and quantify microplastics in household samples.
- Designed a controlled lighting/macro-imaging chamber to classify fibers, fragments, and beads by shape and texture, with exportable particle-count and size-estimation outputs — under ₹4,000 per unit.
- Tested 500+ household samples; reduced testing cost by 95% and improved testing efficiency by 87% versus lab-based methods (FTIR/Raman spectroscopy); endorsed by FSSAI, India's largest food governing body.
- Published in the Indian Patent Journal with a registered Indian Patent by the Intellectual Property Office of India (Application No. IN202641095036 A1, published 14 Aug 2026). Won the CREST Gold Award with 100% marks for uniqueness and viability.

Project CARE: Clinical Anxiety Reduction & Emotional Support System

Jan 2026 – Ongoing

Launched a mobile app on the Play Store with 100+ downloads , App Store coming soon

- Translated clinical-shadowing observations into a web application reducing patient fear during high-stress hospital moments through structured emotional check-ins and workflow explanations.
- Built around a three-stage framework (Capture Fear, Clarify Uncertainty, Create Comfort) supporting communication and emotional readiness without replacing clinical decision-making.

VisionXcelerate — Co-Founder | Wearable Cognitive-Support Glasses for Dementia & Memory Loss

Feb 2024 – Ongoing

Winner, NYAS Junior Academy Wearables Challenge 2024 | Global Semi-Finalist, Diamond Challenge 2026

- Designed an AI-enabled wearable assistive system for dementia care using human-centered clinical research, integrating contextual memory prompts, caregiver-controlled monitoring, GPS localization, and fall detection; building an MVP using Mentra glasses with an integrated caregiver app.
- Synthesized insights from 132+ stakeholders (7 physicians, 100+ institutional caregivers, 25+ family caregivers) to ground the product in real-world patient and caregiver needs.
- Solution validated by neurologists, geriatricians, dementia-care organizations, and AgeTech experts; featured in multiple international policy and innovation publications.

ChoiceX — Co-Founder | AI-Based Career Guidance for Low-Income Students

Nov 2023 – Dec 2025

EY STEM App Challenge Finalist | Life Entrepreneurship Changemaker Award

- Co-founded an AI-integrated career guidance app to reduce youth unemployment among low socio-economic backgrounds in India.
- Built a hybrid recommendation system using TF-IDF vectorization, k-nearest neighbors, and logistic regression to match users with personalized course and job pathways.
- Interviewed 400+ college undergraduates to inform product design; piloted the platform with Diamond School of Management (LOR received).
- Conducted primary/secondary research on the graduate unemployment landscape mentored by EY's STEM App project; won the EY Life Entrepreneurship Changemaker Award (2024).

Summer & Pre-College Programmes

Research Foundations Program — Non-Trivial

Feb 2026 – Apr 2026

- Selected among <1% of global applicants; trained in hypothesis generation, causal inference, literature synthesis, and research design.

Future Doctors I — Harvard Student Agencies

May 2024 – Jul 2024

- Pre-medical immersion covering genetics, immunology, endocrinology, and organic chemistry with Harvard undergraduate mentors.

Advanced Learner's Program (ALP 101 & 201) — ICREST Academy

Apr 2024 – Jul 2024

- Two-week foundational and six-week advanced hands-on program in molecular biology, biochemistry, immunology, and bioinformatics.

Online Courses & Certifications

- Non-Trivial Research Foundations Program — Non-Trivial (Apr 2026)
- Leaf Program — Biology for a Better Tomorrow (Feb 2026)
- Johns Hopkins Cancer Biology Course (Oct 2025)
- Duke University Medical Neuroscience (Aug 2025)
- UChicago Understanding the Brain: The Neurobiology of Everyday Life (Jul 2025)
- Young Member of the New York Academy of Sciences — NYAS (Aug 2024)
- Introduction to Programming Using Python — Udemy (Apr 2024)
- Full Innovation Course — StartupYou (Aug 2023)

Clinical Experience

Clinical Shadowship — Fortis Hospital, under Dr. Ritu Garg

Jun 2026 – Jul 2026

- Shadowed physicians across consultations, inpatient settings, and specialist discussions; observed patient evaluation, treatment planning, and multidisciplinary coordination in a tertiary hospital.
- Rotated across Neurology, Cardiology, and Oncology departments; observed heart transplants, catheterization lab procedures, ECMOs, minimally invasive CABG and stent surgeries, spine and interventional neurology procedures, and chemotherapy.

Clinical Shadowship — Narayana Health

Nov 2025 – Jul 2026

- Rotated across Neurology, Cardiology, and Oncology departments, observing consultations, ward rounds, and surgical procedures; gained exposure to clinical decision-making and patient care systems.
- Identified recurring patterns of patient/caregiver anxiety that directly informed the design of Project CARE.

Positions of Responsibility

- House Captain — Student Council, 10X International School (Oct 2025 – Oct 2026): led 5+ house/school events and competitions, coordinating teams of 100+ students and increasing participation across school activities.
- Class Head for Grades 9 & 10 — represented the student body of 25+ students in Grades 9 & 10, responsible for class coordination and participation in school activities.
- Co-President — Science Club 'The Significant Figures,' 10X International School.
- Editor — Yearbook Committee, 10X International School.
- Innovation Council Member & Hercules Innovation Captain, 10X International School.

Skills & Languages

Programming Languages: Python, R, HTML, CSS, JavaScript

Software & Tools: Figma, OpenCV, TensorFlow Lite, MATLAB

Hardware / Platforms: Raspberry Pi

Soft Skills: Research design & critical inquiry, public speaking, scientific writing, cross-cultural communication, team leadership, stakeholder analysis

English: Full Professional Proficiency

Hindi: Native / Bilingual Proficiency

Telugu: Elementary Proficiency

Korean: Elementary Proficiency

Hobbies & Personal Interests

Guitar, vocals, and dance (award-winning performer in solo and group settings); composes original music. Active in badminton and football (gold medallist in football competitions). State-level swimmer, represented Telangana. Enthusiast of Greek & Roman mythology. Advocate for equitable STEAM access for girls through community outreach.